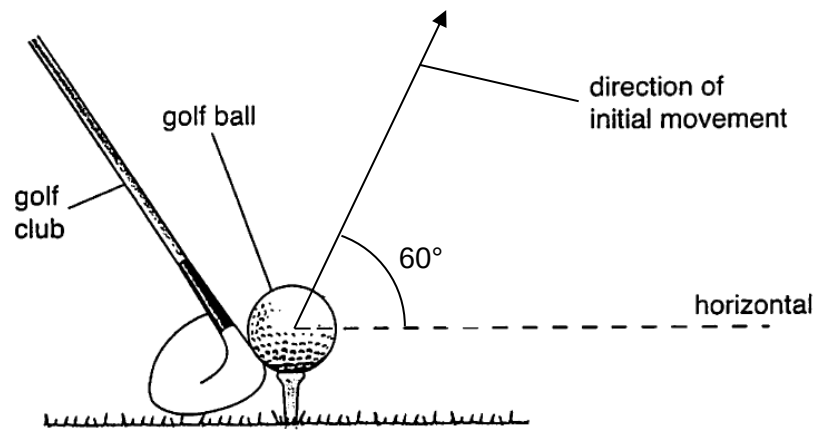
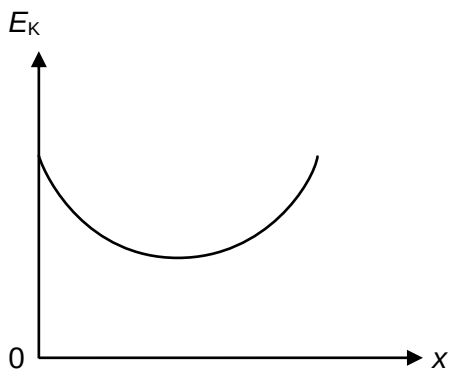


- 7 A golf ball, on level ground, is hit and starts to move at an angle of 60° to the horizontal.

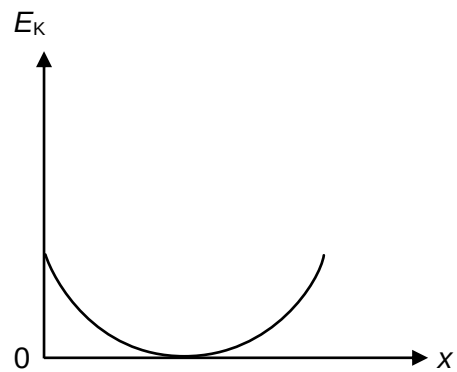


Which graph best represents the variation with horizontal distance x of the kinetic energy E_k of the golf ball? Ignore any effects of air resistance.

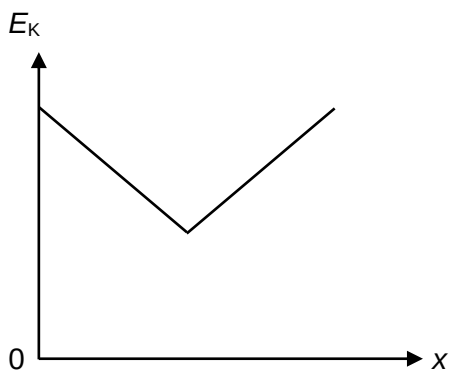
A



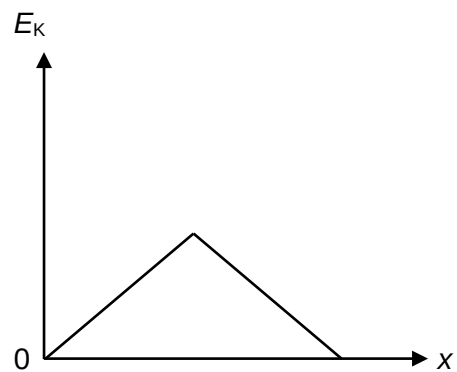
B



C



D



Answer: A

The kinetic energy of the golf ball at the highest point is non-zero. Hence option B is wrong.

The golf ball has the lowest kinetic energy at the highest point. Hence option D is wrong.

The vertical displacement s_y of the golf ball is of the parabolic equation $s_y = u_y t + \frac{1}{2} a_y t^2$ where u_y is the initial vertical velocity, a_y the vertical acceleration of the golf ball and t the time taken for the flight of golf ball. Since $s_x = u_x t$,

$t = \frac{s_x}{u_x} = \frac{x}{u_x}$ where s_x is the horizontal displacement, u_x the initial horizontal velocity of golf ball.

$$\text{Therefore } s_y = u_y \left(\frac{x}{u_x} \right) - \frac{1}{2} g \left(\frac{x}{u_x} \right)^2 = - \left(\frac{g}{2u_x^2} \right) x^2 + \left(\frac{u_y}{u_x} \right) x .$$

Since gravitational potential energy of golf ball of mass m is given as mgs_y , the graph for gravitational potential energy of the golf ball will be a negative parabolic curve.

By principle of conservation of energy, $KE = TE - GPE$, the graph for the kinetic energy of the golf ball will be a positive parabolic curve.